

# NKOLE MARTIN MWANZA

Software Engineer · ERP Developer · Full-Stack Developer

Lusaka, Zambia · +260 962 186 103 · +260 772 145 335 · Martinmwanza24@gmail.com

[linkedin.com/in/nkole-mwanza](https://linkedin.com/in/nkole-mwanza) · [github.com/Nkole101](https://github.com/Nkole101)

## PROFESSIONAL SUMMARY

Final-year Computer Science and Software Engineering student at the University of Zambia (UNZA), graduating November 2026, with hands-on production experience across ERP development, full-stack engineering, and enterprise system administration. As a Software Engineer Intern at Innovative Dynamics Ltd (Nov 2025 – Mar 2026), contributed to ERP deployments serving 3 client organisations and over 800 end users — customising 6 core modules, building 30+ custom DocTypes, and configuring automated approval workflows that reduced procurement cycle times by 40–70% and HR administrative effort by 50–80%. Currently building CCIS, a full-stack parish records management system with a data-driven RBAC engine, as a final-year capstone project, and maintaining Currency Converter, a live Python/FastAPI/React PWA. Also designs and deploys professional portfolio websites for clients end-to-end, from questionnaire to GitHub Pages deployment — 3 live sites delivered to date. Comfortable across frontend, backend, ERP, and DevOps concerns, with a strong foundation in Git, GitHub, CI/CD, and version control workflows. Actively seeking full-time or contract software engineering opportunities.

## PROFESSIONAL EXPERIENCE

### Software Engineer Intern – ERP Developer

Innovative Dynamics Ltd · Automate E ERP Platform

Nov 2025 – Mar 2026

Lusaka, Zambia (On-site)

- Contributed to ERP deployments across 2 client organisations, supporting implementations that onboarded and trained over 400 employees on the Automate E platform.
- Customised 6 core ERP modules — Procurement, Human Resources, Finance & Accounting, Maintenance, Billing, and Payroll — developing client-specific workflows, validation logic, and user interfaces beyond standard platform capabilities.
- Created and configured 30+ custom DocTypes to model new business data structures; reviewed and extended many additional existing DocTypes to meet client requirements.
- Configured end-to-end workflow automation and multi-level approval chains across all major modules, reducing procurement cycle times by an estimated 40–70% and HR administrative effort by 50–80% through elimination of paper-based processes.
- Participated in requirements analysis and ERP configuration sessions with client stakeholders across all 3 organisations, translating business needs into tested, deployable system solutions.
- Implemented server-side Python scripts and JavaScript client scripts to enforce business rules, automate notifications, and improve end-user experience across deployed modules.
- Performed REST API integrations, database configuration, data migration, and query optimisation on MySQL-backed production systems.
- Administered user accounts, role-based access controls, and permission structures across client environments, ensuring correct access segregation aligned to organisational hierarchies.
- Built custom reports, dashboards, and print formats delivering real-time financial and operational visibility — reducing management reporting preparation from days to near real-time.
- Tech: ERPNext · Frappe · Python · JavaScript · MySQL · REST APIs · Workflow Automation · Role-Based Access Control · Custom DocTypes · System Integration

## EDUCATION

### B.Sc. Computer Science & Software Engineering

University of Zambia (UNZA)

Expected Nov 2026

Lusaka, Zambia

- Concentrations: Artificial Intelligence & Modelling · Database Management Systems · Software Development
- Relevant coursework: Data Structures & Algorithms · Operating Systems · Computer Networks · AI · Software Engineering · Web Technologies · Database Management Systems · Computer Architecture
- Final-year capstone project: CCIS — Catholic Community Information System (see Projects).

## PROJECTS

## CCIS — Catholic Community Information System

Final-Year Capstone Project · University of Zambia · In Development

Jan 2026 – Present

[github.com/Nkole101/ccis](https://github.com/Nkole101/ccis)

- Designing and building a production-grade parish records system for St Gabriel Parish — families, SCCs, members, and the full sacramental register.
- Architected a React 19 + Vite frontend with React Router, Recharts dashboards, Framer Motion animations, and light/dark theme support; Express.js REST API backend with PostgreSQL, bcrypt password hashing, and JWT authentication via HttpOnly cookies.
- Designed the full relational database schema across 6 core tables (SCCs, families, members, users, auth tokens, parish register entries); all schema changes managed through versioned migration files.
- Built a data-driven RBAC engine — roles, permissions, and role\_permissions tables define a resource/action grant matrix across 4 roles (superadmin, admin, scc\_leader, member); SCC leaders are scoped to a single community via a user\_scopes table, with route-level middleware enforcing checks on every mutation and sensitive read.
- Tech: React 19 · Vite · Tailwind CSS · Node.js · Express · PostgreSQL · JWT · bcrypt · Axios · Recharts · Framer Motion · Git · GitHub

## Professional Portfolio Websites

Freelance · 3 live sites delivered

2025 – Ongoing

[URLs available on request]

- Designs and builds bespoke single-page professional portfolio websites for individual clients across various professions, managing the complete end-to-end workflow: client questionnaire, content gathering, design, development, and deployment.
- Builds responsive HTML/CSS/JS portfolios covering experience, education, skills, and downloadable documents (CV, transcripts, reference letters).
- Tech: HTML5 · CSS3 · JavaScript · GitHub Pages · Git · GitHub Actions · Responsive Design

## Currency Converter — Python CLI, FastAPI Service & Installable PWA

Personal Project · Live on Vercel & Render

2025 – Present

[github.com/Nkole101/currency-converter-cli](https://github.com/Nkole101/currency-converter-cli)

- Built a shared OOP core (converter.py, api.py, utils.py, logger.py) that powers three independent clients from one codebase: an argparse CLI, a FastAPI REST backend, and a React + Vite PWA frontend.
- Delivers real-time conversion across 150+ currencies with zero configuration, including fuzzy search by currency code, name, or country (e.g. typing “japan” finds JPY).
- Implemented structured logging, input validation, and robust error handling throughout the shared core; deployed the FastAPI backend on Render and the frontend on Vercel.
- Built the frontend as an installable Progressive Web App using vite-plugin-pwa and Workbox service-worker caching — installable on iOS, Android, Windows, and desktop browsers with no app-store dependency.
- Tech: Python · FastAPI · React · Vite · vite-plugin-pwa · Workbox · REST API design · Render · Vercel

## Pharmacy ERP — Full-Stack Inventory & Operations Platform

Personal Project · In Development

2026 – In Progress

- Architecting a full-stack pharmacy inventory and operations platform as a TypeScript monorepo (npm workspaces), with a Prisma + PostgreSQL + Redis API and a Vite + React + TypeScript frontend.
- Implemented JWT access/refresh token authentication and a dedicated background worker process for asynchronous jobs; containerised the full stack — API, worker, PostgreSQL, and Redis — with Docker Compose for reproducible local development.
- Tech: TypeScript · Node.js · Prisma · PostgreSQL · Redis · React · Vite · Docker · Docker Compose · JWT

## CERTIFICATIONS

### CCNA – Introduction to Networks

Cisco Networking Academy · 2025

Network protocols, IP addressing, routing and switching fundamentals, and network security concepts.

### IT Essentials – PC Hardware and Software

Cisco Networking Academy

PC assembly, hardware configuration, software installation, troubleshooting, and preventive maintenance.

### NDG Linux Essentials & NDG Linux Unhatched

Cisco / NDG

Linux command line, file system navigation, user and permission management, and shell scripting fundamentals.

## TECHNICAL SKILLS

---

**Languages:** Python · JavaScript · TypeScript · Java · SQL · Kotlin · HTML5 · CSS3 · Bash

**Frontend:** React 19 · Vite · Tailwind CSS · React Router · Recharts · Framer Motion · Lucide React · Responsive CSS · PWA / Service Workers · Workbox

**Backend:** Node.js · Express.js · FastAPI · Frappe · ERPNext · Custom DocTypes · Workflow Automation · REST API design · JWT · bcrypt · HttpOnly cookies

**Databases:** PostgreSQL · MySQL · SQLite · Redis · Prisma · SQLAlchemy · data migration · query optimisation

**DevOps & Version Control:** Git · GitHub · GitHub Pages · GitHub Actions (CI/CD) · branch-based deployment · Docker · Docker Compose · Nginx · XAMPP

**Networking & OS:** TCP/IP · DNS · DHCP · routing/switching (CCNA) · Windows · Linux (Ubuntu, CLI) · shell scripting

**Tools:** VS Code · PyCharm · IntelliJ IDEA · Android Studio · MySQL Workbench · ProjectLibre · Chrome DevTools · Microsoft Office Suite

**Soft Skills:** Team leadership · stakeholder communication · requirements analysis · technical documentation · agile/iterative development · client training

## REFERENCES

---

*Available on request.*